Generating-function exercises

1. Derive the OGF of $a_n=n$ by multiplication or differentiation, explaining a_0.
2. If $a_n=n+1$ and $b_n=1$ for all n, find the coefficient of x^4 in A(x)B(x). Relate it to a prefix sum.
3. Use generating functions to show that nonnegative solutions of $x_1+2x_2+3x_3=n$ are counted by the degree-n coefficient of $1/((1-x)(1-x^2)(1-x^3))$.
4. Find the OGF and closed coefficient formula for partitions of n into at most two positive unordered parts.
5. Two distinguishable component types each contain zero or one labeled atom. Find their EGFs and product. Distinguish the ordinary x^2 coefficient from the structure count on two specified labels.
6. For $A(x)=\sum a_n x^n$, express the OGF of $t_n = n\sum_{k=0}^{n} a_k$ in a form ready for coefficient extraction.
7. Find $[x^n]x^2/(1-3x)^3$, distinguishing n=0,1 from $n\geq 2$.
8. In formal series, compute the inverse of $A(x)=2-3x+x^2$ through degree three and check by multiplication.
9. Let $E(x)=e^x-1$. Explain and evaluate $4![x^4]E(x)^3/3!$. Also give the ordinary x^4 coefficient and explain the separate roles of 3! and 4!.
10. Prove that odd-part and distinct-part partition families are equinumerous. List the five partitions on each side for n=7.
11. Allocate n interchangeable weekly work units to three job types consuming 1,2,3 units, with unused units allowed. Let a_n count allocations consuming at most n. Introduce slack, derive the OGF, find a_7, and verify by direct enumeration. Explain why a three-denominator product alone does not count the inequality.
12. Prove the SET construction exp(Â(x)) for nonempty labeled components. Explain division by k! and the condition Â(0)=0. Count partitions of four labels into two unlabeled nonempty blocks, giving each ordered-size contribution.
13. Apply the odd-to-distinct bijection and its inverse to $26=5^3+3^2+1^5$, where exponents denote multiplicities. Prove Ferrers conjugation is an involution using $\lambda'_j = \#\{i : \lambda_i \geq j\}$, and calculate it for 4+2+1.
14. With at most two A's, two B's, and one C, find word counts at lengths three and four, listing multiplicities and the EGF. For length-four words over {A,B,C} with even A-count, check the EGF answer by direct classification.

Generating-function exercises (continued)

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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